Assignment 8

1. System definition

AI hiring assistant analyses the candidates and then chooses suitable hires based on applicant-submitted materials, historical data and organisational criteria. It will turn candidates into a set of scores and weights to analyse the applicants as objectively as possible. In the output we can get c concrete scoring on parameters chosen within the system to access whether and which candidate are hire worthy.

2. Bias and Fairness Synthesis

Bias is a general tendency or prejudice that affects a persons judgement. If the person is aware of it it is explicit and if not it is implicit. Biases may be positive and negative, intentional or unintentional. Bias does not have specific targets.
Fairness on the other hand is an attempt to mitigate biases.
Implicit Bias in its nature is not deliberate and the main difficulty arises in its determination since for the average person their biases seem fair. Where implicit bias can be more accurately identified is through the source, which can be culture, media, societal exposure, stereotyping. It can be a belief, habit disposition or more.
Algorithms may be not entirely fair due to implicit biases that exist within them. Whether through an imperfect database, lack of balanced exposure or subjectivity of fairness itself - these biases will affect our systems.
I chose AI hiring assistant as the system I will investigate. In it the biases are encountered through the systems preferences. For example search engines can introduce biases. The AI algorithms by their nature are opaque and there is no way of proving why the algorithm ranked one candidate over the other. The applicants data as well is subject to analysis and therefore can be accessed by unintended individuals whether through a data leak or else.

3. Deployment Scenarios

Deployment 1 An AI Hiring Assistant is used by HR recruiters and hiring managers in a large private company to screen job applicants. The system analyzes resumes, cover letters, and assessments to generate rankings and recommendations, but the **final hiring decision remains with human recruiters**, making it an advisory tool. The level of consequence is high, as its recommendations influence employment opportunities, income, and career progression. This baseline deployment highlights concerns about implicit and algorithmic bias because recruiters may rely heavily on AI-generated rankings, even though they retain decision-making authority.

Deployment 2: The same AI Hiring Assistant is deployed in a government agency to recruit civil servants. In this context, the user group changes from private-sector recruiters to government HR officers, and the institutional context changes from a private company to the public sector. The system also shifts from being advisory to decision-making, automatically rejecting applicants who do not meet a predefined score before any human review. The level of consequence becomes very high, as decisions affect access to public employment and may have legal implications. Compared with the baseline, the increased authority of the AI and the public-sector setting make transparency, accountability, explainability, and fairness much more significant because individuals have stronger expectations of equal treatment and procedural justice.

Deployment 3: The AI Hiring Assistant is repurposed by a university career services office to help students improve their resumes before applying for jobs. Instead of being used by employers to select candidates, the system provides feedback on resume quality, identifies missing skills, and suggests improvements. It remains an advisory system, and the level of consequence is low to medium because it does not directly determine employment outcomes but may indirectly influence future job success. Compared with the baseline, the user group changes from recruiters to students, the institutional context changes from employment to education, and the system's purpose shifts from selection to preparation. Analytically, this matters because the ethical concerns move away from discriminatory hiring decisions toward issues of equal access to AI-assisted career support, the standardization of resumes, and whether students are encouraged to optimize their applications for AI systems rather than authentically representing their abilities.

4. Mediation Pathway

Pathway State	**Transformation / Mediation**	**Potential Bias**	Bias Type	Bias Origin
Source	Applicants submit resumes, cover letters, and assessments.	Applicants may have unequal access to education, resume-writing support, or professional experience.	Sampling bias; representation bias	Fresh
Vector	Documents are converted into structured features (skills, education, experience).	Important contextual information may be ignored, while proxies such as university prestige or employment gaps may be overemphasized.	Proxy variable bias; feature selection bias	Fresh
Boundary Object	Structured applicant profiles are transferred from applicants to the AI system.	Translation into standardized variables can oversimplify diverse experiences.	Abstraction bias	Emergent
Vector	AI compares applicants with historical hiring data and organizational criteria.	Historical hiring patterns may reflect past discrimination.	Historical bias; labeling bias	Inherited
Decision Point	AI generates rankings and recommendations.	The optimization model may prioritize majority-group characteristics or maximize efficiency over fairness.	Algorithmic bias; optimization bias	Emergent
Boundary Object	Rankings are presented through the recruiter interface.	Ranking order and score presentation may encourage overreliance on AI outputs.	Interface framing; authority bias	Fesh
Destinatio n	Recruiters review recommendations and make hiring decisions.	Recruiters may defer excessively to AI recommendations (automation bias).	Authority assumption confirmation bias	Emergent
Final Conseque nce	Candidate is hired or rejected.	Systematic exclusion of particular demographic groups.	Feedback loop	Emergent

Pathway Stage	Transformation / Mediation	Potential Bias	Bias Type	Bias Origin
Source	Applicants submit standardized government applications.	Certain communities may have fewer opportunities to meet eligibility requirements.	Representation bias	Fresh
Vector	Applications are transformed into standardized evaluation scores.	Public-sector criteria may unintentionally disadvantage some groups.	Feature selection bias	Fresh
Boundary Object	Applicant information enters the government AI screening system.	Standardization may remove contextual information relevant to disadvantaged applicants.	Abstraction bias	Emergent
Vector	AI applies historical recruitment models and government scoring rules.	Historical recruitment practices may reproduce institutional discrimination.	Historical bias	Inherited
Decision Point	AI automatically rejects applicants below a threshold.	Small prediction errors become final decisions because no human review occurs.	Threshold bias; automation bias	Emergent
Boundary Object	Rejection decisions are communicated to HR and applicants.	Limited explanation reduces transparency and opportunities to appeal.	Explainability bias	Fresh
Destination	HR officers implement AI decisions.	Human oversight is minimal, increasing dependence on automated outputs.	Authority assumption	Emergent
Final Consequence	Applicants are accepted or rejected for public employment.	Errors directly affect access to public-sector jobs and equal opportunity.	Institutional feedback loop	Emergent

Pathway Stage	Transformation / Mediation	Potential Bias	Bias Type	Bias Origin
Source	Students upload resumes and career information.	Students have varying levels of prior experience and access to career preparation resources.	Representation bias	Fresh
Vector	AI extracts resume features and compares them with successful job applications.	Historical hiring patterns may define what an "ideal" resume looks like.	Historical bias	Inherited
Boundary Object	AI-generated feedback is presented to students and career advisors.	Recommendations may promote standardized resume styles over diverse experiences.	Framing bias	Emergent
Decision Point	Students decide whether to follow AI recommendations.	Students may assume AI advice is always correct.	Authority assumption	Fresh
Boundary Object	Revised resumes are submitted to employers.	Students who lack access to AI coaching may be disadvantaged.	Access bias	Fresh
Destination	Employers evaluate improved resumes.	AI-assisted applicants may appear more competitive regardless of actual ability.	Competitive advantage bias	Emergent
Final consequence	Students gain improved employment prospects.	Unequal access to AI support may widen existing educational inequalities.	Opportunity bias	Emergent

Comparative Analysis

Bias locations consistent across all three deployments

The following bias locations appear in every deployment regardless of context:

- Source: Representation or sampling bias due to unequal applicant backgrounds and opportunities.
- Feature extraction (Vector): Bias introduced when rich human experiences are reduced to structured variables or proxy measures.
- Boundary object: Abstraction bias occurs whenever human identities are translated into machine-readable data.

- Historical model (Vector): Inherited bias from historical hiring or evaluation data.
- Decision point: Algorithmic outputs may privilege certain groups depending on model design and optimisation objectives.
- Human interaction (Destination): Users may over-trust AI recommendations due to automation bias or authority assumptions

Bias locations specific to particular deployments

- Private company (Deployment 1): Interface framing bias is particularly important because recruiters interpret ranked candidate lists and may rely heavily on presentation order.
- Government recruitment (Deployment 2): Threshold bias, explainability bias, and institutional feedback loops become much more significant because the AI directly determines eligibility for public employment with limited human oversight.
- University career services (Deployment 3): Access bias and competitive advantage bias emerge because not all students have equal access to AI-assisted resume coaching, creating disparities before the hiring process even begins.

New bias risks created by changing the deployment context

Changing the deployment context introduces new forms of bias that are absent or less significant in the baseline

1)Moving from private recruitment to government hiring increases the consequences of automation. Because the AI becomes a decision-maker rather than an advisor, threshold bias, reduced transparency, limited appeal mechanisms, and procedural fairness become critical concerns.

2) Moving from employer recruitment to university career services changes the system's purpose from selecting applicants to preparing them. This creates new risks related to unequal access to AI support, the standardisation of resumes, and incentives for students to optimise applications for AI screening rather than accurately representing their skills.

Overall, while representation bias, historical bias, abstraction bias, and automation bias persist across all deployments, the institutional context, decision authority, and consequence level determine which additional biases emerge and how significant their impacts become

5. Decision Pathway Analysis

Deployment 1: Private Company (Baseline)

The AI produces candidate rankings and recommendations, which are reviewed by HR recruiters who make the final hiring decision. Human judgment enters at the final stage. If the AI is biased, qualified candidates may be unfairly rejected. Accountability is shared between recruiters, the company, and the AI provider. Recruiters mainly see AI-generated scores rather than applicants' full backgrounds.

Deployment 2: Government Recruitment

The AI produces eligibility scores that are used by government HR officers to automatically reject some applicants. Human judgment is largely removed from the screening stage. If the AI is biased, applicants may lose access to public jobs unfairly. Accountability is more diffuse, and decision-makers rely mainly on AI scores instead of applicants' complete profiles.

Deployment 3: University Career Services

The AI provides resume feedback and improvement suggestions to students and career advisors. Human judgment remains throughout, as students choose whether to follow the advice. Biased recommendations may disadvantage students or encourage standardized resumes. Accountability is shared, and employers ultimately see AI-optimized resumes rather than students' original applications.

6. Harm Analysis

Direct harm: The AI Hiring Assistant can directly harm individuals by incorrectly ranking or rejecting qualified applicants due to biased or inaccurate outputs. In the university deployment, poor recommendations may also reduce the quality of students' job applications.

Indirect harm: The system can reinforce existing inequalities over time. For example, if biased hiring decisions repeatedly favor certain groups, future hiring data becomes biased as well, creating a feedback loop that further disadvantages underrepresented applicants. In the university setting, students without access to AI career support may become less competitive.

Distribution of harm: Harms are not evenly distributed. They are more likely to affect women, ethnic minorities, people with disabilities, older workers, and applicants from less prestigious educational or socioeconomic backgrounds. Employers and well-represented groups are generally less affected and may even benefit from the system. These harms align with the course concepts of discrimination, unfairness, exclusion, unequal opportunity, and feedback loops, as the system can amplify existing social inequalities rather than reduce them

7. Critical Reflection

Responsibility for bias-related harm cannot be placed solely on the AI system, the people using it, or the surrounding institution. Instead, bias emerges from the interaction between historical data, system design, organizational policies, and human decision-making. The AI reflects patterns found in its training data, while organizations determine how much authority the system is given and how its outputs are used. Likewise, human decision-makers influence whether AI recommendations are questioned or accepted without sufficient scrutiny.

The level of responsibility also changes across the three deployments. In the private company, recruiters retain the final decision, so human judgment remains an important safeguard. In the government deployment, where the AI has greater decision-making authority, the institution has a stronger responsibility to ensure fairness, transparency, and accountability. In the university deployment, the AI only provides advice, but unequal access to the technology can still create unfair advantages. Overall, reducing bias requires shared responsibility among AI developers, organizations, and users, together with regular auditing, transparent processes, and meaningful human oversight.